Green Bonds EDA (Improved Treatment)

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```
# Load in all libraries
library(dplyr)
library(tidyverse)
library(readxl)
library(ggplot2)
library(maps)
library(tidyr)
```

```
# Solar Data
solar_2024 <- read_excel("Solar.xlsx", sheet = "2024")
solar_2023 <- read_excel("Solar.xlsx", sheet = "2023")
solar_2022 <- read_excel("Solar.xlsx", sheet = "2022")
solar_2021 <- read_excel("Solar.xlsx", sheet = "2021")
solar_2020 <- read_excel("Solar.xlsx", sheet = "2020")
solar_2019 <- read_excel("Solar.xlsx", sheet = "2019")
solar_2018 <- read_excel("Solar.xlsx", sheet = "2018")
solar_2017 <- read_excel("Solar.xlsx", sheet = "2017")
solar_2016 <- read_excel("Solar.xlsx", sheet = "2016")
solar_2015 <- read_excel("Solar.xlsx", sheet = "2015")
solar_2014 <- read_excel("Solar.xlsx", sheet = "2014")
solar_2013 <- read_excel("Solar.xlsx", sheet = "2013")

# Wind Data
wind_2024 <- read_excel("Wind.xlsx", sheet = "2024")
wind_2023 <- read_excel("Wind.xlsx", sheet = "2023")
wind_2022 <- read_excel("Wind.xlsx", sheet = "2022")
wind_2021 <- read_excel("Wind.xlsx", sheet = "2021")
wind_2020 <- read_excel("Wind.xlsx", sheet = "2020")
wind_2019 <- read_excel("Wind.xlsx", sheet = "2019")
wind_2018 <- read_excel("Wind.xlsx", sheet = "2018")
wind_2017 <- read_excel("Wind.xlsx", sheet = "2017")
wind_2016 <- read_excel("Wind.xlsx", sheet = "2016")
wind_2015 <- read_excel("Wind.xlsx", sheet = "2015")
wind_2014 <- read_excel("Wind.xlsx", sheet = "2014")
wind_2013 <- read_excel("Wind.xlsx", sheet = "2013")

# Green Bond Issuance and RPS Data
GBI <- read_excel("GB_RPS Hard Code.xlsx")

# Pre-Trends Wind Data
pre_wind <- read_csv("Wind Pre.csv")
```

```r
# Pre-Trends Solar Data
pre_solar <- read_excel("Solar 2010 - 2012.xlsx")

# Real GDP Data
gdp_raw <- read_csv("SAGDP1_ALL_AREAS_1997_2025.csv")

# State Population Data (2010 - 2020)
intercensal_raw <- read_excel(
  "nst-est2020int-pop.xlsx",
  sheet = 1,
  skip = 3
)

# State Population Data (2020 - 2024)
vintage2024_raw <- read.csv("NST-EST2024-POPCHG2020-2024.csv", stringsAsFactors = FALSE)
```

```r
# Cleaning Pre-Wind Data

clean_wind_panel <- pre_wind %>%

# Remove any rows where the operational year is missing
filter(!is.na(p_year)) %>%

# Select only the state, year, and capacity columns
select(t_state, p_year, t_cap) %>%

# Group by state and year, and convert individual turbine kW to total MW added
group_by(t_state, p_year) %>%
summarize(mw_added = sum(t_cap, na.rm = TRUE) / 1000, .groups = 'drop') %>%

# Open 50-state grid for pre-trends window
complete(t_state, p_year = 2000:2026, fill = list(mw_added = 0)) %>%
arrange(t_state, p_year) %>%
group_by(t_state) %>%

# Cumulative sum holds capacity at 0 until first project,
# then holds total capacity flat forward through the years
mutate(total_wind_mw = cumsum(mw_added)) %>%
rename(state = t_state, year = p_year) %>%

# Filter the outcome variable to strictly include 2010-2012
filter(year >= 2010 & year <= 2013)

# Split by each year and pivot year column to display state and capacity by year
clean_wind_panel <- clean_wind_panel %>%
  select(state, year, total_wind_mw) %>%
  group_by(year) %>%
  group_split(.keep = FALSE)
wind_2010 <- clean_wind_panel[[1]]
wind_2011 <- clean_wind_panel[[2]]
wind_2012 <- clean_wind_panel[[3]]

# Used to compare discrepancies between two different data sources for wind capacity
```

```r
wind_2013_test <- clean_wind_panel[[4]]
```

```r
# Cleaning Pre-Solar Data

# 2. Extract state, and 2010 - 2013 capacity data
solar_wide_cumulative <- pre_solar %>%
  select(
    State = 1,
    raw_2010 = 2,
    raw_2011 = 3,
    raw_2012 = 4,
    raw_2013 = 5
  )

# Convert from solar capacity added to cumulative capacity
solar_wide_cumulative <- solar_wide_cumulative %>%
  mutate(
    # 2010 is just the base year
    Year_2010 = replace_na(raw_2010, 0),

    # 2011 = 2010 + 2011
    Year_2011 =
      replace_na(raw_2010, 0) +
      replace_na(raw_2011, 0),

    # 2012 = 2010 + 2011 + 2012
    Year_2012 =
      replace_na(raw_2010, 0) +
      replace_na(raw_2011, 0) +
      replace_na(raw_2012, 0),

    # 2013 = 2010 + 2011 + 2012 + 2013
    # Used to compare discrepancies between two different data sources for solar capacity
    Year_2013_test =
      replace_na(raw_2010, 0) +
      replace_na(raw_2011, 0) +
      replace_na(raw_2012, 0) +
      replace_na(raw_2013, 0)
  ) %>%

  # 5. Output clean state name and cumulative columns for each year
  select(State, Year_2010, Year_2011, Year_2012, Year_2013_test)
```

```r
# Cleaning Post-Wind Data

# Add year columns
wind_2010 <- wind_2010 %>%
  rename(State = `state`, wind_capacity = `total_wind_mw`) %>%
  mutate(year = 2010) %>%
  arrange(State)

wind_2011 <- wind_2011 %>%
  rename(State = `state`, wind_capacity = `total_wind_mw`) %>%
```

```r
  mutate(year = 2011) %>%
  arrange(State)

wind_2012 <- wind_2012 %>%
  rename(State = `state`, wind_capacity = `total_wind_mw`) %>%
  mutate(year = 2012) %>%
  arrange(State)

# Loop through every year from 2013 to 2024
for (y in 2013:2024) {

  # Generate the variable name ("wind_year")
  var_name <- paste0("wind_", y)

  # Temporary data frame to do cleaning
  temp_df <- get(var_name)
  cleaned_df <- temp_df %>%

    # Adds the current loop's year
    mutate(year = y) %>%
    rename(wind_capacity = `Nameplate Capacity (MW)`) %>%
    group_by(State, year) %>%
    summarize(
      wind_capacity = sum(wind_capacity, na.rm = TRUE), .groups = "drop"
      ) %>%
    arrange(State)

  # 4. Save the cleaned data frame back to its original variable name
  assign(var_name, cleaned_df)
}

# Combine all years (2010 - 2024)
post_wind_all = bind_rows(
  wind_2010, wind_2011, wind_2012, wind_2013, wind_2014,
  wind_2015, wind_2016, wind_2017, wind_2018, wind_2019,
  wind_2020, wind_2021, wind_2022, wind_2023, wind_2024
  )

# To account for discrepancies between pre and post data sources
# Scale pre-trend data
wind_2013_uswtdb <- wind_2013_test %>%
  rename(State = state, wind_uswtdb = total_wind_mw)

wind_ratios <- wind_2013 %>%
  rename(wind_eia = wind_capacity) %>%
  left_join(wind_2013_uswtdb, by = "State") %>%
  mutate(

    # Ratio = EIA Capacity / USWTDB Capacity
    scale_factor = wind_eia / wind_uswtdb
  ) %>%
  mutate(scale_factor = if_else(
    is.nan(scale_factor) | is.infinite(scale_factor), 1, scale_factor)
```

```
    ) %>%
  select(State, scale_factor)

# Scale down 2010-2012 wind capacities to meet the EIA baseline
post_wind_all <- post_wind_all %>%
  left_join(wind_ratios, by = "State") %>%
  mutate(
    wind_capacity = if_else(
      year <= 2012, wind_capacity * scale_factor, wind_capacity
      )
  ) %>%
  select(-scale_factor)

# Create a master data frame containing all 48 contiguous states
# Excludes Hawaii and Alaska
contiguous_48 <- state.abb[!state.abb %in% c("AK", "HI")]
master_states <- data.frame(State = contiguous_48)

# Pivot wind data from long format to wide format
wind_wide <- post_wind_all %>%
  pivot_wider(
    names_from = year,
    values_from = wind_capacity,
    names_prefix = "yr_"
  )

# Left join the master list with wind data and sort states alphabetically
final_wind_panel <- master_states %>%
  left_join(wind_wide, by = "State") %>%
  arrange(State)

final_wind_panel_clean <- final_wind_panel %>%

  # Temporarily bring columns back to a long format to work row-by-row per state
  pivot_longer(
    cols = starts_with("yr_"),
    names_to = "year",
    values_to = "wind_capacity"
  ) %>%
  # 2. Group by State and sort by year to ensure sequential order
  group_by(State) %>%
  arrange(State, year) %>%
  # Fill NA values with the previous year's value
  fill(wind_capacity, .direction = "down") %>%
  # If it's still NA (meaning it never had any capacity), replace with 0
  mutate(wind_capacity = replace_na(wind_capacity, 0)) %>%
  # Bring it back to the wide layout
  pivot_wider(
    names_from = year,
    values_from = wind_capacity
  ) %>%
  ungroup()
```

```r
# Cleaning Post-Solar Data

# Add year columns
solar_2010 <- solar_wide_cumulative %>%
  select(State, solar_capacity = Year_2010) %>%
  mutate(
    year = 2010,
    State = state.abb[match(State, state.name)]
    ) %>%
  arrange(State)

solar_2011 <- solar_wide_cumulative %>%
  select(State, solar_capacity = Year_2011) %>%
  mutate(
    year = 2011,
    State = state.abb[match(State, state.name)]) %>%
  arrange(State)

solar_2012 <- solar_wide_cumulative %>%
  select(State, solar_capacity = Year_2012) %>%
  mutate(
    year = 2012,
    State = state.abb[match(State, state.name)]) %>%
  arrange(State)

solar_2013_test <- solar_wide_cumulative %>%
  select(State, solar_capacity = Year_2013_test) %>%
  mutate(
    year = 2013,
    State = state.abb[match(State, state.name)]) %>%
  arrange(State)

# Loop through every year from 2013 to 2024
# Same way we did for the post wind data
for (y in 2013:2024) {
  var_name <- paste0("solar_", y)
  temp_df <- get(var_name)
  cleaned_df <- temp_df %>%
    mutate(year = y) %>%
    rename(solar_capacity = `Nameplate Capacity (MW)`) %>%
    group_by(State, year) %>%
    summarize(
      solar_capacity = sum(solar_capacity, na.rm = TRUE), .groups = "drop"
      ) %>%
    arrange(State)
  assign(var_name, cleaned_df)
}

post_solar_all = bind_rows(
  solar_2010, solar_2011, solar_2012, solar_2013,
  solar_2014, solar_2015, solar_2016, solar_2017,
  solar_2018, solar_2019, solar_2020, solar_2021,
  solar_2022, solar_2023, solar_2024
```

```
  )

# Isolate the 2013 Berkeley Lab data and convert names to abbreviations
solar_2013_lbnl_check <- solar_wide_cumulative %>%
  select(State, solar_lbnl = Year_2013_test) %>%
  mutate(
    State = state.abb[match(State, state.name)]
  )

# Join the EIA 2013 data with the Berkeley Lab data
solar_ratios <- solar_2013 %>%
  rename(solar_eia = solar_capacity) %>%
  left_join(solar_2013_lbnl_check, by = "State") %>%
  mutate(

    # Create a conditional scale factor to prevent division by zero or Na errors
    # If there's missing or 0 data, don't scale by ratio to avoid dividing by 0
    # Otherwise, scale by the ratio of the two data source values
    scale_factor = if_else(
      is.na(solar_eia) | is.na(solar_lbnl) | solar_eia == 0 | solar_lbnl == 0,
      1,
      solar_eia / solar_lbnl
    )
  ) %>%
  select(State, scale_factor)

# Apply the ratio to the pre-trends years (2010-2012)
post_solar_all <- post_solar_all %>%
  left_join(solar_ratios, by = "State") %>%
  mutate(
    # Double check protection for any lingering NAs
    scale_factor = replace_na(scale_factor, 1),
    solar_capacity = if_else(
      year <= 2012, solar_capacity * scale_factor, solar_capacity
      )
  ) %>%
  select(-scale_factor)

contiguous_48 <- state.abb[!state.abb %in% c("AK", "HI")]
master_states <- data.frame(State = contiguous_48)
solar_wide <- post_solar_all %>%
  group_by(State, year) %>%
  summarize(solar_capacity = sum(solar_capacity, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = year,
    values_from = solar_capacity,
    names_prefix = "yr_"
  )

final_solar_panel <- master_states %>%
  left_join(solar_wide, by = "State") %>%
  arrange(State)
```

```r
# Interpolate missing values
final_solar_panel_clean <- final_solar_panel %>%
  pivot_longer(
    cols = starts_with("yr_"),
    names_to = "year",
    values_to = "solar_capacity"
  ) %>%
  group_by(State) %>%
  arrange(State, year) %>%
  fill(solar_capacity, .direction = "down") %>%
  mutate(solar_capacity = replace_na(solar_capacity, 0)) %>%
  pivot_wider(
    names_from = year,
    values_from = solar_capacity
  ) %>%
  ungroup()
```

```r
# Merge state population data: 2010-2019 (Intercensal)
# + 2020-2024 (Vintage 2024)

# Rename columns: first column is geography, next is 2010 base, then 2011-2019,
# then 2020 census
colnames(intercensal_raw) <- c(
  "Geography", "Base2010", paste0("Y", 2010:2019), "Census2020"
)

intercensal_states <- intercensal_raw %>%
  filter(grepl("^\\.", Geography)) %>%
  mutate(State = sub("^\\.", "", Geography)) %>%
  filter(State != "Puerto Rico") %>%

  # filter(State != "District of Columbia") %>%
  select(State, paste0("Y", 2010:2019)) %>%
  pivot_longer(
    cols = paste0("Y", 2010:2019),
    names_to = "Year",
    values_to = "Population"
  ) %>%
  mutate(Year = as.integer(sub("Y", "", Year)))


vintage2024_states <- vintage2024_raw %>%
  filter(trimws(as.character(SUMLEV)) %in% c("40", "040")) %>%
  filter(NAME != "Puerto Rico") %>%
  select(NAME, POPESTIMATE2020, POPESTIMATE2021, POPESTIMATE2022,
                POPESTIMATE2023, POPESTIMATE2024) %>%
  rename(State = NAME) %>%
  pivot_longer(
    cols = starts_with("POPESTIMATE"),
    names_to = "Year",
    values_to = "Population"
  ) %>%
  mutate(Year = as.integer(sub("POPESTIMATE", "", Year)))
```

```r
# Merge into one long-format table: 2010-2024, states only
state_population_2010_2024 <- dplyr::bind_rows(
  intercensal_states, vintage2024_states
  ) %>%
  arrange(State, Year)

# Reshape to wide format: one row per state, one column per year
state_population_wide <- state_population_2010_2024 %>%
  pivot_wider(
    names_from = Year, values_from = Population, names_prefix = "Y"
    )

# Map full state names to 2-letter postal codes
population_long <- state_population_2010_2024 %>%
  filter(State %in% c(state.name, "District of Columbia")) %>%
  mutate(
    year = as.numeric(Year),
    Population = as.numeric(Population),
    State = case_when(
      State == "District of Columbia" ~ "DC",
      TRUE ~ state.abb[match(State, state.name)]
    )
  ) %>%
  select(State, year, Population)

# Save outputs
write.csv(
  state_population_2010_2024,
  "state_population_2010_2024_long.csv",
  row.names = FALSE
  )

write.csv(
  state_population_wide,
  "state_population_2010_2024_wide.csv",
  row.names = FALSE
  )
```

```r
# Cleaning Real GDP data

gdp_long <- gdp_raw %>%
  # Keep only the Real GDP rows (LineCode is 1)
  filter(LineCode == 1) %>%

  # Drop the "United States" row and any regional summaries
  filter(GeoName %in% c(state.name, "District of Columbia")) %>%

  # 4. Keep only the State name column and years 2010 - 2024
  select(State = GeoName, `2010`:`2024`) %>%

  # Reshape from wide (columns) to long (rows) to match population data
  pivot_longer(
    cols = `2010`:`2024`,
```

```
    names_to = "year",
    values_to = "real_gdp"
  ) %>%

  # Fix data types, scale by 1 million,
  # and convert full state names to 2-letter postal codes
  mutate(
    year = as.numeric(year),
    real_gdp = as.numeric(real_gdp) * 1000000,
    State = case_when(
      State == "District of Columbia" ~ "DC",
      TRUE ~ state.abb[match(State, state.name)]
    )
  )
```

```
# Building the Panel data set

# Create a dedicated long-format version for regressions and plotting
final_wind_panel_long <- final_wind_panel %>%

# Remove any rows where the operational year is missing/blank
# Pivot to long format
 pivot_longer(
   cols = starts_with("yr_"),
   names_to = "year",
   values_to = "wind_capacity"
 ) %>%

# Strip "yr_" so the year column becomes a clean, parse-able number
 mutate(year = as.numeric(gsub("yr_", "", year))) %>%

# Sort sequentially by state and year to prepare for the carry-forward step
 group_by(State) %>%
 arrange(State, year) %>%

# Handle missing data gaps using Last Observation Carried Forward (LOCF)
 fill(wind_capacity, .direction = "down") %>%

# Switch any remaining early NAs to 0
 mutate(wind_capacity = replace_na(wind_capacity, 0)) %>%

# Ungroup so it behaves like a standard data frame
 ungroup()

# Create a dedicated long-format version for solar regressions and plotting
final_solar_panel_long <- final_solar_panel %>%
 pivot_longer(
   cols = starts_with("yr_"),
   names_to = "year",
   values_to = "solar_capacity"
 ) %>%
 mutate(year = as.numeric(gsub("yr_", "", year))) %>%
 group_by(State) %>%
```

```
 arrange(State, year) %>%
 fill(solar_capacity, .direction = "down") %>%
 mutate(solar_capacity = replace_na(solar_capacity, 0)) %>%
 ungroup()

# One-hot-encoding of corporate & municipal issuance, and RPS issuance
# Corporate: 1 if current year >= corporate year, otherwise 0
# Municipal: 1 if current year >= municipal year, otherwise 0
# RPS: 1 if current year >= RPS enacted AND current year < RPS expired
# If never expired, remains 1
# GB: 1 if a state issued either type of green bond,
# 0 if a state issued neither
policy_clean <- GBI %>%

# 1. Standardize state names to postal abbreviations
 mutate(State = state.abb[match(State, state.name)]) %>%
 filter(!is.na(State)) %>%

# 2. Convert character strings/NA values to proper numeric years or infinity
 mutate(
   corp_start = as.numeric(gsub("N/A", NA, `Corporate Year`)),
   muni_start = as.numeric(gsub("N/A", NA, `Municipal Year`)),
   rps_start = as.numeric(gsub("N/A", NA, `RPS Year`)),
   rps_end = as.numeric(gsub("N/A", NA, `RPS Repealed/Expired`)),

   # If RPS was never repealed, set its expiration to infinity
   rps_end = if_else(is.na(rps_end), Inf, rps_end)
 ) %>%
 select(State, corp_start, muni_start, rps_start, rps_end)

# Merge Wind and Solar long panels into a single joint panel
joint_energy_panel <- final_wind_panel_long %>%
 full_join(final_solar_panel_long, by = c("State", "year"))

# Combine panels and compute the logical thresholds
final_joint_regression_panel <- final_wind_panel_long %>%

# Join wind and solar together side-by-side
 full_join(final_solar_panel_long, by = c("State", "year")) %>%

# Join cleaned timeline parameters
 left_join(policy_clean, by = "State") %>%

# Join population metrics sequentially
 left_join(population_long, by = c("State", "year")) %>%

# Join BEA real GDP metrics sequentially
 left_join(gdp_long, by = c("State", "year")) %>%

# Build the binary metrics dynamically
 mutate(
   is_corporate = if_else(!is.na(corp_start) & year >= corp_start, 1, 0),
   is_public = if_else(!is.na(muni_start) & year >= muni_start, 1, 0),
```

```
    gb = if_else(is_corporate == 1 | is_public == 1, 1, 0),

    # RPS Policy Dummy
    is_rps = if_else(!is.na(rps_start) & year >= rps_start & year < rps_end, 1, 0)
  ) %>%

# Calculate all per capita and economic density metrics
  mutate(
    real_gdp_per_capita       = real_gdp / Population,
    wind_capacity_per_capita  = wind_capacity / Population,
    solar_capacity_per_capita = solar_capacity / Population
  ) %>%

# Drop the following columns to only include the one-hot-encodings in the panel
  select(-corp_start, -muni_start, -rps_start, -rps_end)

# Export final panel data set to a CSV file
# This is the data set to be used in Stata for our econometric analysis
write.csv(final_joint_regression_panel, "panel.csv", row.names = FALSE)
```

```
# Improved Treatment: Technology-Specific Green Bond Financing

# Use-of-proceeds audit of the appendix (Table 6): one row per issuer,
# with what the proceeds funded and where the projects are located
gb_uop <- read_csv("green_bond_use_of_proceeds.csv", show_col_types = FALSE) %>%
  mutate(state = state.abb[match(state, state.name)])

# 1. Issuances whose proceeds funded solar and/or wind generation
tech_rows <- gb_uop %>%
  filter(proceeds_to %in% c("Solar", "Wind", "Both")) %>%
  mutate(tech = if_else(proceeds_to == "Both", "solar,wind", tolower(proceeds_to))) %>%
  separate_rows(tech, sep = ",") %>%
  select(state, tech, bond_year = year_table, entity) %>%

  # 2. Coding decisions (documented in the CSV's project_location and flag columns)
  # Drop bonds with no in-state capacity: Alabama Power bought out-of-state wind,
  # Forbright lends nationwide
  filter(!entity %in% c("Alabama Power (Southern Co.)", "Forbright Bank")) %>%
  mutate(

    # Code by project location, not issuer state
    # TEP (AZ issuer) funded Oso Grande wind in New Mexico
    # SPS (NM issuer) funded Hale wind in Texas
    state = case_when(
      entity == "Tucson Electric Power Co."       ~ "NM",
      entity == "Southwestern Public Service Co." ~ "TX",
      TRUE ~ state
    ),

    # One consistent first year for the multistate NSP-Minnesota issuer
    bond_year = if_else(state %in% c("MN", "ND", "SD") & tech == "wind",
                        as.numeric(params$nsp_first_year), as.numeric(bond_year)),
```

```
    # Medium-confidence codings (project-level allocation not confirmed)
    medium = (state %in% c("OK", "ND", "SD") & tech == "wind") |
             (state == "MS" & tech == "solar")
  ) %>%

  # Multistate solar portfolios also treat these states:
  # Duke Energy Carolinas (NC and SC), Dominion Energy (VA, NC, SC and CA)
  bind_rows(tribble(
    ~state, ~tech,   ~bond_year, ~entity,                 ~medium,
    "SC",   "solar", 2018,       "Duke Energy Carolinas", FALSE,
    "NC",   "solar", 2018,       "Dominion Energy",       FALSE,
    "SC",   "solar", 2018,       "Dominion Energy",       FALSE,
    "CA",   "solar", 2018,       "Dominion Energy",       TRUE
  ))

# Robustness option: drop medium-confidence codings
if (!params$include_medium) {
  tech_rows <- tech_rows %>% filter(!medium)
}

# 3. Placebo: verified or issuer-type "Other" bonds (no solar or wind),
# excluding O-I (not an Illinois issuance) and NRG (not a green bond)
other_rows <- gb_uop %>%
  filter(proceeds_to %in% c("Other", "Other (hydro)"),
         confidence != "Not verified",
         !entity %in% c("Owens-Illinois Inc.", "NRG Energy / Texas Issuers")) %>%
  mutate(
    # Brookfield's White Pine hydro bond was issued in 2017, not 2019
    bond_year = if_else(entity == "Brookfield Renewable Partners", 2017, as.numeric(year_table)),
    tech = "other"
  ) %>%
  select(state, tech, bond_year)

# First treatment year by state and technology (0 = never treated)
tech_cohorts <- bind_rows(tech_rows, other_rows) %>%
  group_by(State = state, tech) %>%
  summarize(g = min(bond_year), .groups = "drop") %>%
  pivot_wider(names_from = tech, values_from = g, names_prefix = "g_")

# Corrected baseline: original market-entry coding with documented errors fixed
# IL corporate: O-I's 2019 bond was a euro issue by a Dutch subsidiary; O-I is HQ'd in Ohio
# TX corporate: NRG's 2020 deal was a sustainability-linked bond, not a green bond
# MN, ND, SD corporate: same NSP-Minnesota issuer, so one consistent first year
policy_fix <- policy_clean %>%
  mutate(
    corp_start = case_when(
      State %in% c("IL", "TX") ~ NA_real_,
      State %in% c("MN", "ND", "SD") ~ as.numeric(params$nsp_first_year),
      TRUE ~ corp_start
    )
  ) %>%
  select(State, corp_fix = corp_start, muni_fix = muni_start)
```

```r
panel_improved <- final_joint_regression_panel %>%
  left_join(tech_cohorts, by = "State") %>%
  left_join(policy_fix, by = "State") %>%
  mutate(

    # Technology-specific cohorts and absorbing treatment dummies
    across(c(g_solar, g_wind, g_other), ~ coalesce(as.numeric(.x), 0)),
    gb_solar = if_else(g_solar > 0 & year >= g_solar, 1, 0),
    gb_wind  = if_else(g_wind  > 0 & year >= g_wind,  1, 0),

    # Placebo: green bonds that funded only non-solar, non-wind uses
    gb_other = if_else(g_other > 0 & year >= g_other, 1, 0),
    placebo_sample = if_else(g_solar == 0 & g_wind == 0, 1, 0),

    # Corrected baseline dummies
    is_corporate_fix = if_else(!is.na(corp_fix) & year >= corp_fix, 1, 0),
    is_public_fix    = if_else(!is.na(muni_fix) & year >= muni_fix, 1, 0),
    gb_fix           = if_else(is_corporate_fix == 1 | is_public_fix == 1, 1, 0),
    g_gb_fix         = coalesce(pmin(corp_fix, muni_fix, na.rm = TRUE), 0),
    g_corp_fix       = coalesce(corp_fix, 0),
    g_muni_fix       = coalesce(muni_fix, 0),

    # Outcomes in watts per person (Table 1 definition) and log transforms
    solar_w_pc  = solar_capacity_per_capita * params$watts_scale,
    wind_w_pc   = wind_capacity_per_capita  * params$watts_scale,
    ln_solar_pc = log(solar_w_pc + 1),
    ln_wind_pc  = log(wind_w_pc + 1),
    ln_gdp_pc   = log(real_gdp_per_capita),

    # Numeric state identifier for did::att_gt()
    state_id = as.integer(factor(State))
  ) %>%
  select(-corp_fix, -muni_fix) %>%
  arrange(State, year)
```

```r
# Checks on panel_improved

# Balanced panel: 48 states x 15 years, one row per state-year
stopifnot(
  nrow(panel_improved) == 48 * 15,
  !any(duplicated(panel_improved[c("State", "year")]))
)

# Every treatment dummy is absorbing (never switches from 1 back to 0)
absorbing <- panel_improved %>%
  group_by(State) %>%
  summarize(across(c(gb_solar, gb_wind, gb_other, gb_fix), ~ all(diff(.x) >= 0)),
            .groups = "drop")
stopifnot(all(absorbing[-1]))

# Cohort sizes under each treatment definition
cohort_table <- function(gvar) {
  panel_improved %>%
```

```
    distinct(State, g = .data[[gvar]]) %>%
    count(g, name = "states") %>%
    mutate(g = if_else(g == 0, "Never treated", as.character(g)))
}
knitr::kable(cohort_table("g_wind"),  caption = "Wind treatment cohorts")
```

Table 1: Wind treatment cohorts

g	states
Never treated	34
2016	1
2017	1
2018	2
2019	4
2020	2
2021	3
2022	1

```
knitr::kable(cohort_table("g_solar"), caption = "Solar treatment cohorts")
```

Table 2: Solar treatment cohorts

g	states
Never treated	36
2016	1
2018	5
2019	3
2021	3

```
knitr::kable(cohort_table("g_other"), caption = "Placebo treatment cohorts")
```

Table 3: Placebo treatment cohorts

g	states
Never treated	26
2013	1
2014	2
2015	1
2016	1
2017	4
2019	3
2020	4
2021	2
2022	2
2023	2

```r
# State-years where the corrected baseline differs from the original gb
panel_improved %>%
  filter(gb != gb_fix | is_corporate != is_corporate_fix) %>%
  count(State, name = "state_years_changed") %>%
  knitr::kable(caption = "Changes from original to corrected baseline coding")
```

Table 4: Changes from original to corrected baseline coding

State	state_years_changed
IL	6
MN	2
ND	1
SD	1
TX	5

```r
# Export improved panel for Stata and R
# - Drop the original market-entry dummies (is_corporate, is_public, gb), which
#   contain the coding errors, and keep the corrected versions under the
#   original names so existing Stata code runs unchanged
# - Drop the cohort-year columns (g_*); every treatment variable exported is 0/1
#   (Stata rebuilds first-treatment years with egen, as in JoE_Stata.do)
panel_improved_export <- panel_improved %>%
  select(-is_corporate, -is_public, -gb, -starts_with("g_")) %>%
  rename(is_corporate = is_corporate_fix,
         is_public    = is_public_fix,
         gb           = gb_fix) %>%
  select(State, year, wind_capacity, solar_capacity, Population, real_gdp,
         is_corporate, is_public, gb, gb_wind, gb_solar, gb_other,
         placebo_sample, is_rps, everything())

# Every treatment variable must be strictly 0 or 1
treatment_vars <- c("is_corporate", "is_public", "gb", "gb_wind", "gb_solar",
                    "gb_other", "placebo_sample", "is_rps")
stopifnot(all(sapply(panel_improved_export[treatment_vars],
                     function(x) all(x %in% c(0, 1)))))

write.csv(panel_improved_export, "panel_improved.csv", row.names = FALSE)
```

```r
# Plotting helpers shared by all figures

# Line chart of capacity per capita: thin lines for each state,
# thick lines for the average of treated vs never-treated states
plot_capacity_lines <- function(outcome, gvar, tech, file) {

  line_data <- panel_improved %>%
    mutate(group = if_else(.data[[gvar]] > 0,
                           paste(tech, "green bond financing"),
                           paste("Never treated for", tolower(tech)))) %>%
    select(State, year, group, value = all_of(outcome))

  group_means <- line_data %>%
```

```
    group_by(year, group) %>%
    summarize(value = mean(value), .groups = "drop")

  p <- ggplot() +
    geom_line(data = line_data,
              aes(x = year, y = value, group = State, color = group),
              linewidth = 0.4, alpha = 0.35) +
    geom_line(data = group_means, aes(x = year, y = value, color = group),
              linewidth = 1.4) +
    scale_color_manual(
      values = setNames(c("#C07A5A", "gray55"),
                        c(paste(tech, "green bond financing"),
                          paste("Never treated for", tolower(tech)))),
      name = NULL
    ) +
    scale_x_continuous(breaks = seq(2010, 2024, by = 2)) +
    labs(
      title = paste(tech, "Capacity Per Capita Over Time For Lower 48 States (2010 - 2024)"),
      subtitle = "Thin lines are individual states; thick lines are group means",
      x = "Year",
      y = paste(tech, "Capacity Per Capita (W / Person)")
    ) +
    theme_minimal() +
    theme(
      legend.position = "bottom",
      plot.title = element_text(hjust = 0.5, face = "bold"),
      plot.subtitle = element_text(hjust = 0.5, size = 10, color = "gray30")
    )

  print(p)
  ggsave(filename = file, plot = p, width = 10, height = 6, dpi = 300)
}

# Choropleth of 2024 capacity per capita, labeling the treated states
plot_capacity_map <- function(outcome, gvar, tech, file) {

  map_2024 <- panel_improved %>%
    filter(year == 2024) %>%
    select(State, capacity = all_of(outcome)) %>%
    mutate(region = tolower(state.name[match(State, state.abb)]))

  geo_data <- map_data("state") %>%
    left_join(map_2024, by = "region")

  treated_states <- panel_improved %>%
    distinct(State, g = .data[[gvar]]) %>%
    filter(g > 0) %>%
    pull(State)

  treated_labels <- data.frame(
    State = state.abb,
    long = state.center$x,
    lat = state.center$y
```

```
  ) %>%
    filter(!State %in% c("AK", "HI"), State %in% treated_states)

  p <- ggplot() +
    geom_polygon(data = geo_data,
                 aes(x = long, y = lat, group = group, fill = capacity),
                 color = "white", linewidth = 0.2) +
    geom_text(data = treated_labels, aes(x = long, y = lat, label = State),
              color = "black", fontface = "bold", size = 3.5, check_overlap = TRUE) +
    scale_fill_gradientn(
      name = "W / Person",
      colors = c("#E6D3B3", "#D49A73", "#C07A5A"),
      labels = scales::comma_format(accuracy = 0.1),
      na.value = "grey90"
    ) +
    coord_map("albers", lat0 = 39, lat1 = 45) +
    labs(
      title = paste("Cumulative US", tech, "Capacity Per Capita by State (2024)"),
      subtitle = paste("Labeled states received green bond financing for in-state",
                       tolower(tech), "capacity")
    ) +
    theme_void() +
    theme(
      legend.position = "right",
      legend.box.margin = margin(t = 0, r = 20, b = 0, l = 0),
      plot.title = element_text(hjust = 0.5, face = "bold"),
      plot.subtitle = element_text(hjust = 0.5, size = 10, color = "gray30")
    )

  print(p)
  ggsave(filename = file, plot = p, width = 10, height = 6, dpi = 300)
}

# Stacked timeline of first treatment year, one label per state
plot_timeline <- function(gvar, title, color, file) {

  timeline_data <- panel_improved %>%
    distinct(State, year = .data[[gvar]]) %>%
    filter(year > 0) %>%
    arrange(year, State) %>%
    group_by(year) %>%
    mutate(stack_index = row_number()) %>%
    ungroup()

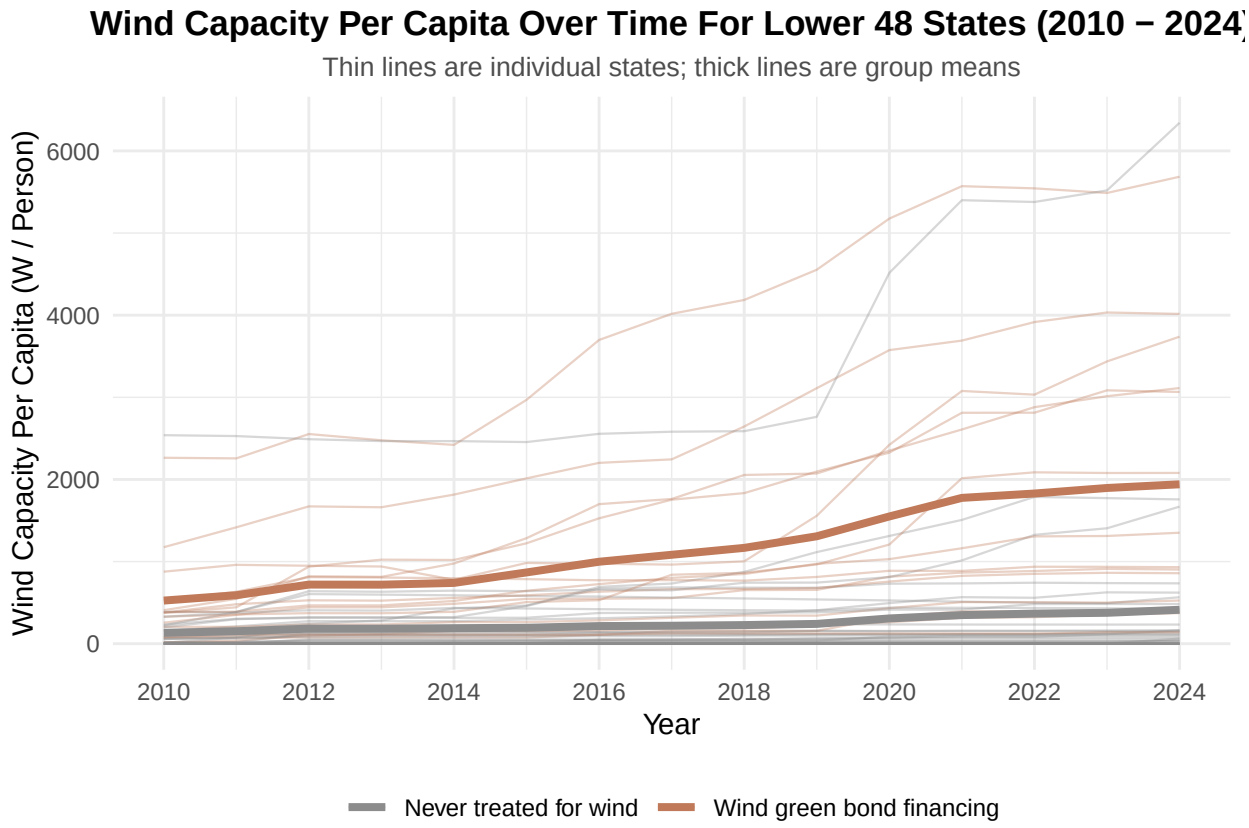
  p <- ggplot(timeline_data, aes(x = year, y = stack_index, label = State)) +
    geom_hline(yintercept = 0, color = "gray50", linewidth = 0.5) +
    geom_text(color = color, fontface = "bold", size = 4.5) +
    scale_x_continuous(breaks = seq(2013, 2024, by = 1), limits = c(2012.5, 2024.5)) +
    scale_y_continuous(expand = expansion(mult = c(0.1, 0.2))) +
    labs(title = title, x = "Year of First Qualifying Issuance", y = "") +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
```

```
      panel.grid.major.y = element_blank(),
      panel.grid.minor.y = element_blank(),
      axis.text.y = element_blank(),
      axis.ticks.y = element_blank()
    )

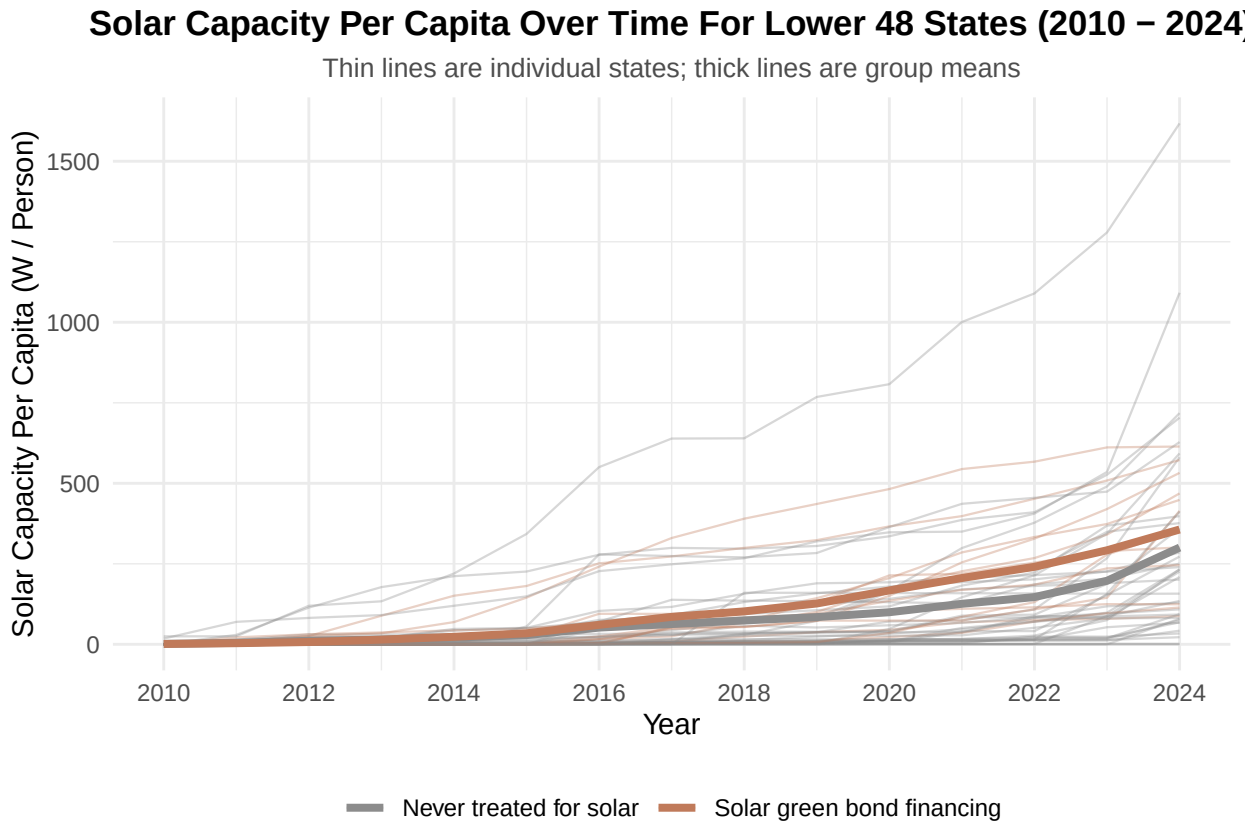
  print(p)
  ggsave(filename = file, plot = p, width = 10, height = 5, dpi = 300)
}
```

```
# Wind and Solar Line Charts (treated vs never-treated)

plot_capacity_lines("wind_w_pc",  "g_wind",  "Wind",  "wind_capacity_per_capita_timeline.png")
```

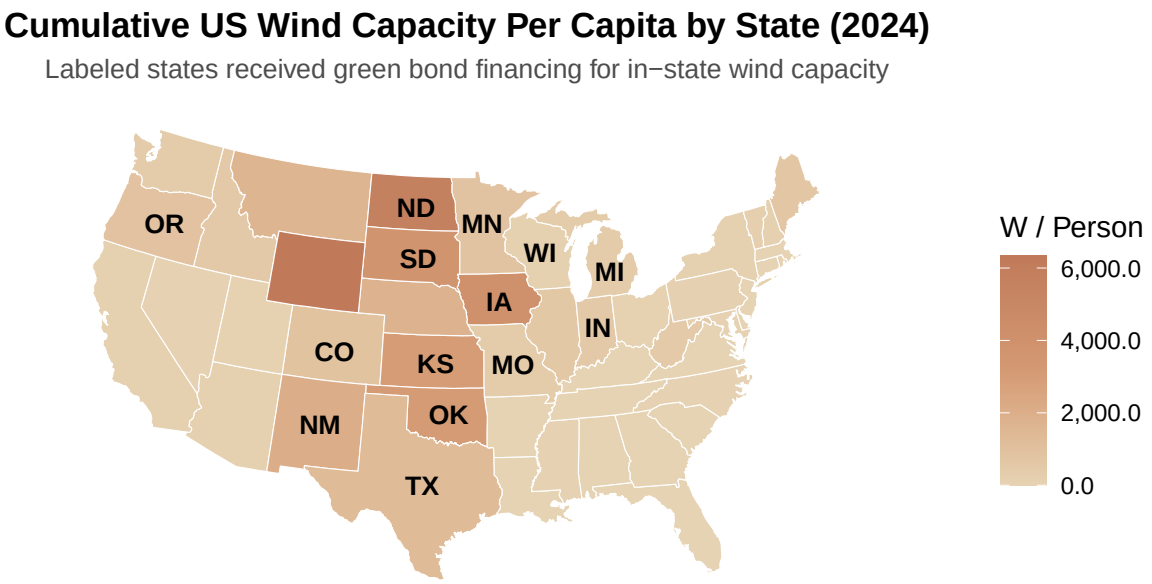



```
plot_capacity_lines("solar_w_pc", "g_solar", "Solar", "solar_capacity_per_capita_timeline.png")
```

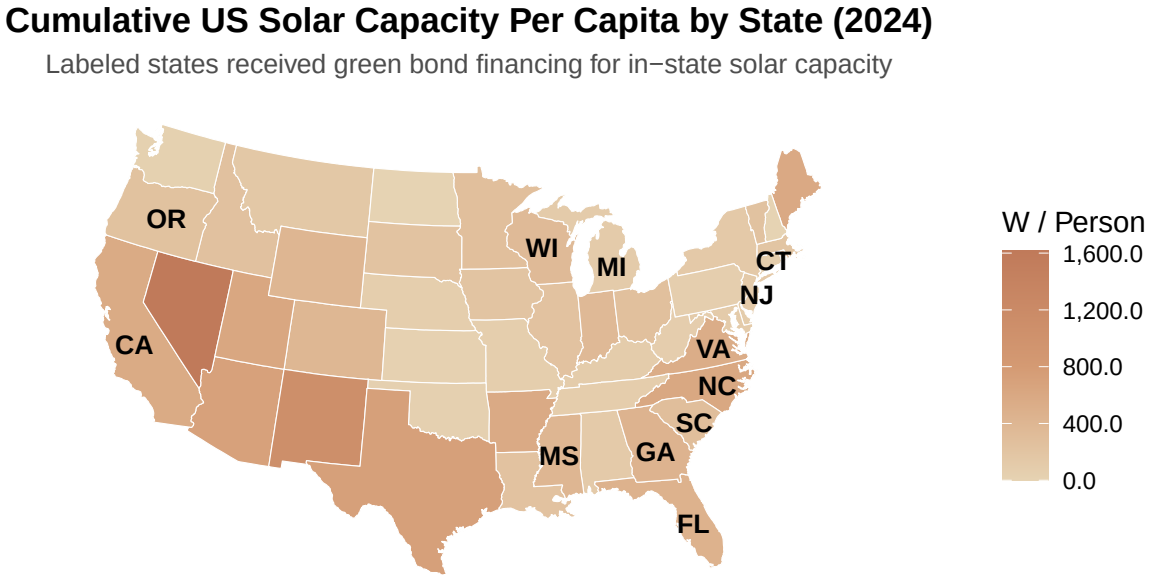



```
# Wind and Solar Maps (labels = technology-specific treated states)

plot_capacity_map("wind_w_pc",  "g_wind",  "Wind",  "wind_map_per_capita.png")
```

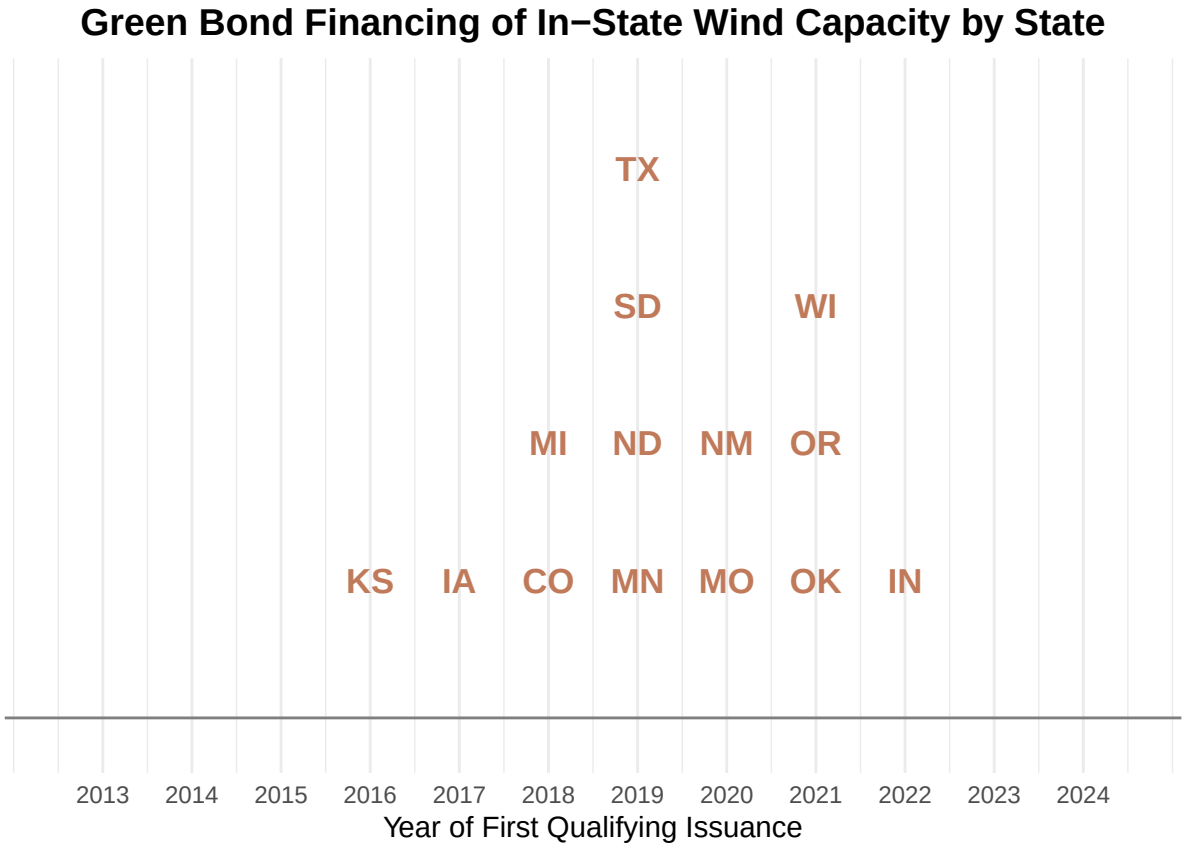



```
plot_capacity_map("solar_w_pc", "g_solar", "Solar", "solar_map_per_capita.png")
```

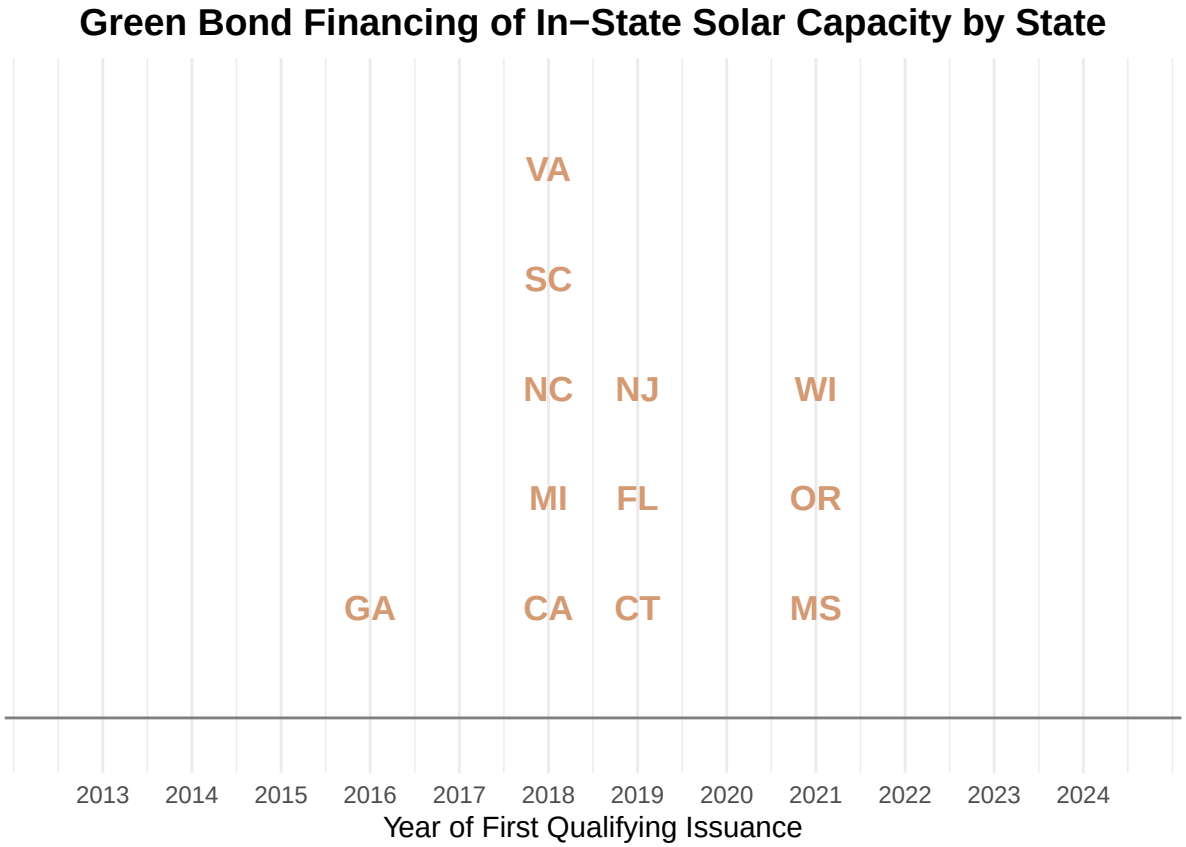



```
# Treatment Timelines

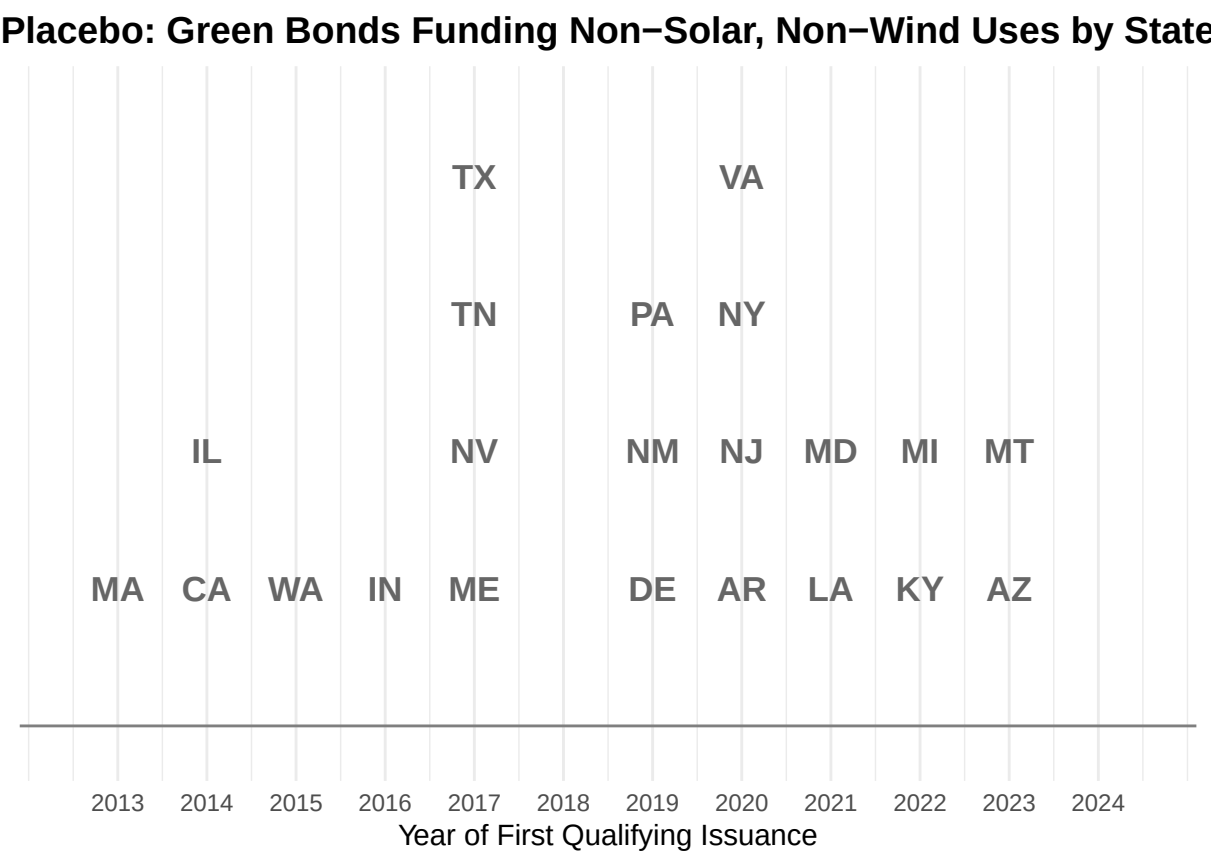
# Main treatment: first green bond funding in-state wind / solar capacity
plot_timeline("g_wind",  "Green Bond Financing of In-State Wind Capacity by State",
              "#C07A5A", "wind_treatment_timeline.png")
```




```
plot_timeline("g_solar", "Green Bond Financing of In-State Solar Capacity by State",
              "#D49A73", "solar_treatment_timeline.png")
```




```
# Placebo treatment: green bonds funding only non-solar, non-wind uses
plot_timeline("g_other", "Placebo: Green Bonds Funding Non-Solar, Non-Wind Uses by State",
              "gray40", "placebo_treatment_timeline.png")
```


Placebo: Green Bonds Funding Non−Solar, Non−Wind Uses by State
TX
VA
TN
PA
NY
IL
NV
NM
NJ
MD
MI
MT
MA
CA
WA
IN
ME
DE
AR
LA
KY
AZ
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
Year of First Qualifying Issuance